U.PORTO

FACULDADE DE CIÊNCIAS
UNIVERSIDADE DO PORTO

Analysis and forecast of the Earth Surface temperature

Time series and Forecasting

José Pedro Gama de Sousa Rodrigues
up202006455

Data Science

2023/2024

Index

Introduction

Methods and Results

Exploratory Data Analysis

Forecasting

- SARIMA
- Exponential Smoothing
- Cross-Validation

Conclusion

References

Attachments

Introduction

Global warming is a problem that gets increasingly concerning every year. One way of analysing this issue is through the measurement of land temperatures and evaluating its difference throughout the years. In this report my objective is to analyse a time series concerning this topic, and forecast the land temperature of the Earth in the next coming years with the most accuracy possible. My time Series was found in a CSV in kaggle ("Climate Change: Earth Surface Temperature Data (kaggle.com)") and contained 9 columns with the only relevant information being in 2 (Date and LandAverageTemperature).

Method and Results

The CSV used was "GlobalTemperatures.csv" from Kaggle. The program language used in this project was Python with jupyter notebook as the IDE. Six main libraries were used for a variety of purposes, these were pandas, numpy, matplot, seaborn, statsmodels and sklearn. With the main focus being in statsmodels that was helpful in the production of specific forecasting models such as SARIMA or Exponential Smoothing.

Exploratory Data Analysis (EDA)

The main components in performing an EDA of a time-series consists of loading the data, pre-processing the data and analysing the stationarity. After loading the data through the command "pd.read_csv()", the pre-processing of the data consisted of handling missing values and dropping unnecessary columns or rows. In this specific case only 'Date' and 'LandAverageTemperature' were necessary, so 6 columns were removed and the remaining column was renamed into 'Temperature' for further clarity and simplicity. The dropped rows with missing values lead to the data being shortened a high amount from the data starting in 1750 to now starting in 1850, this is likely due to the high uncertainty in the initial part of the data. The visualization of the data was made using matplot and seaborn library.

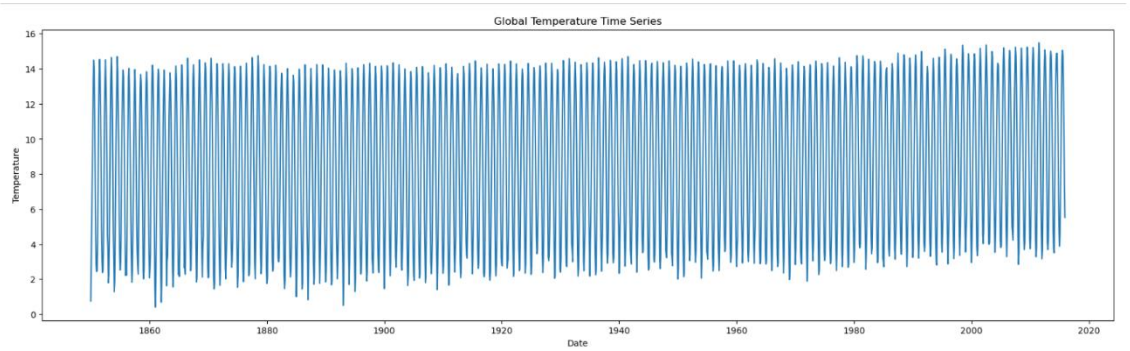


Figure 1.Time series of CSV (GlobalTemperatures.csv)

Initial analysis of the plot indicated strong seasonality and a slight increase in trend. This suspicions were made clearer through the making of further plots. These plots involved Yearly

temperatures focusing on specific months, a lag plot of 12 different lags and another yearly average temperature to check for trend. Both the lag plot and fig(2) showed seasonality. The lag plots had great correlation at different lags especially at lag 12 indicating yearly seasonality, fig.(2) demonstrated that every year had similar temperatures with a specific distribution (month of July having the highest temperatures). Fig.(3) showed a clear upward trend on yearly average temperatures.

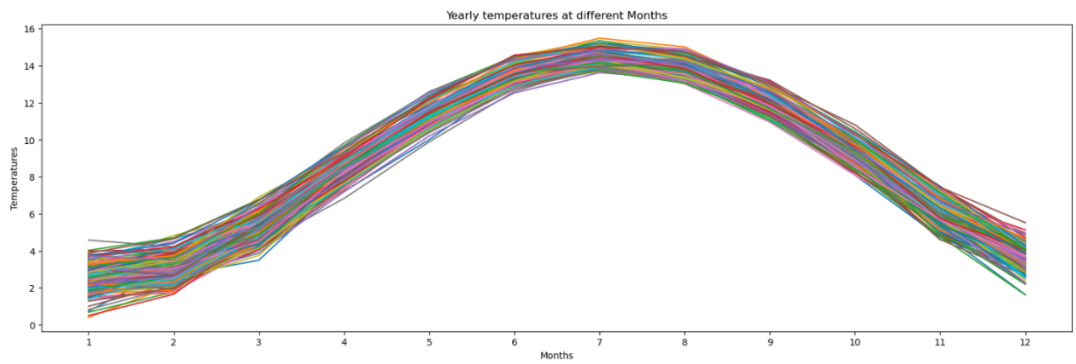


Figure 2. Yearly temperatures at different Months

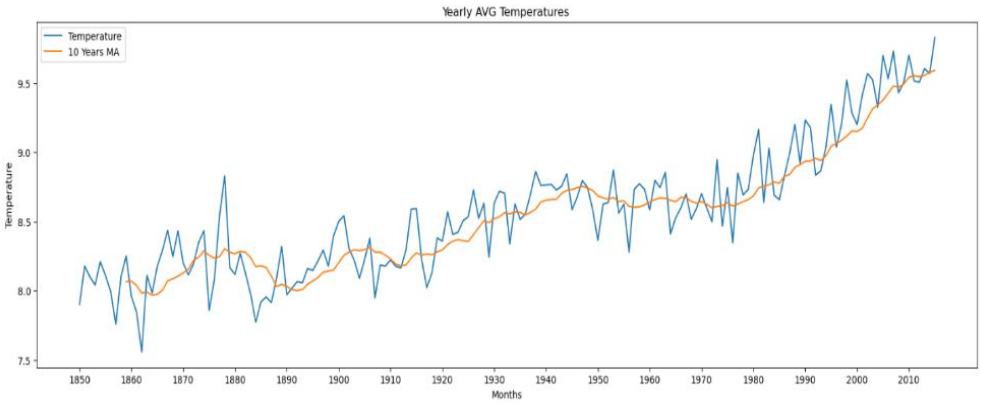


Figure 3.Yearly average temperatures

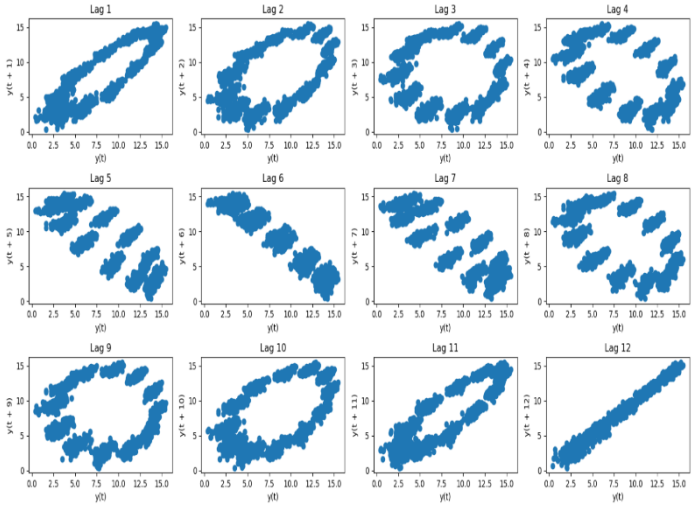


Figure 4. Lag plots of the original time series

In order to have better analysis, two methods of decomposition were performed. The first one used the seasonal_decomposition import from statsmodel. This decomposition model would separated the Original time series in 4 plots: Trend, Seasonality, Residuals and Original (for reference). A lag plot was then made using the residuals of the data in order to calculate stationarity. The lag plot of the residual showed no trend nor seasonality demonstrating the usefulness of this decomposition method.

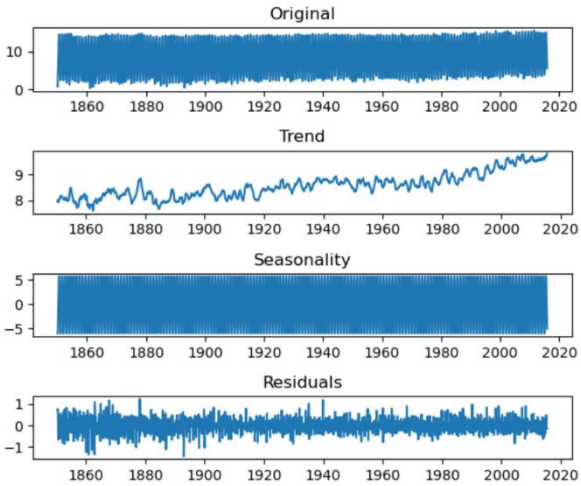


Figure 5. Time series decomposition with 4 plots consisting of original, trend, seasonality and residuals

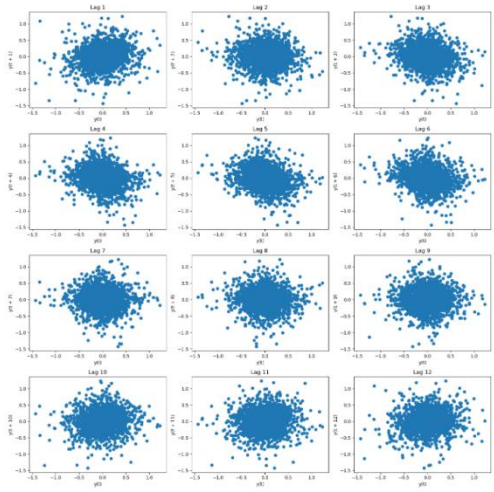


Figure 6. Lag plots of Residuals from seasonal decomposition

The other decomposition method that was used was the filtering of data with a difference operator. In the case of this time series the seasonal difference operator used was $\nabla S=1-B^{12}$ resulting in a stationary residual with no trend nor seasonality. A further difference was investigated but was dropped due to the suspicion of over differencing, since it could lead to misleading results.

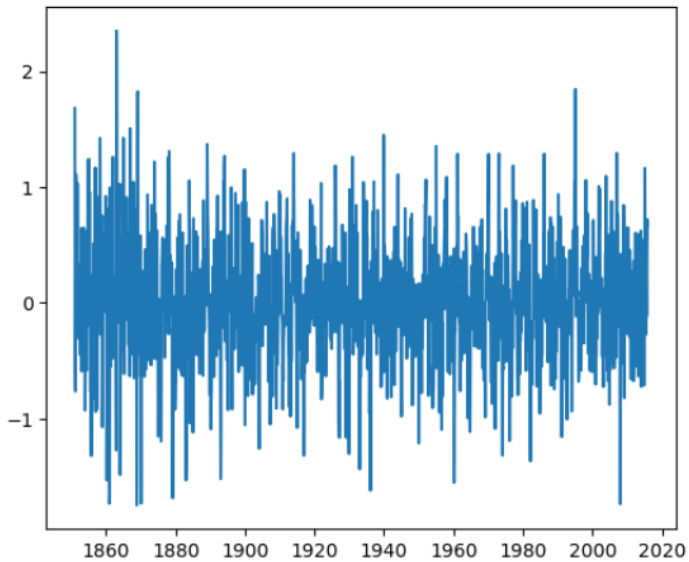


Figure 7. Time Series plot after using a Seasonal difference operator

To confirm the stationarity, the augmented Dickey-Fuller test(ADF) and the Kwiatkowski-Phillips-Schmidt-Shin(KPSS) test were done. The ADF determined the presence of a unit root in the series, the null hypothesis of the series being that it has a unit root and indicating non-stationarity. The KPSS is the opposite with respect to ADF, the null hypothesis indicating stationarity.

Table 1. Stationarity evaluation using ADF and KPSS comparing the original data, the residual from seasonal decomposition and the plot after a seasonal difference operator

	ADF			KPSS		
	Original	∇^{12}	Residual	Original	∇^{12}	Residual
Test Statistic	-1.46	-1.28e+01	-1.75 e+01	2.33	0.027	0.29
P-value	0.56	7.33e-24	4.00 e-30	0.01	0.10	0.10
#Lags Used	26	2.4e+01	2.4e+01	55	18	1238
Nº of Observations	1965	1.96e+03	1.96e+03	0.35	0.35	0.35
Critical Value (1%)	-3.43	-3.43e+00	-3.43e+00	0.46	0.46	0.46
Critical Value (5%)	-2.86	-2.86e+00	-2.86e+00	0.57	0.57	0.57
Critical Value (10%)	-2.57	-2.57e+00	-2.57e+00	0.73	0.74	0.73

Analysing table.1 the original timeseries fails to reject the null hypothesis of the ADF and rejects the null hypothesis of the KPSS by having a p-value lower than 0.5 and a test Statistic higher than the critical values. This means that the original time series is not stationary. Both the residual of the seasonal_decompose and the seasonal difference operator have an ADF test statistic lower than the critical value and the p-value extremely small. For the KPSS the test statistic is also lower than the critical value and the p-value is greater than 0.05 meaning that they are both stationary.

Forecasting

SARIMA

SARIMA stands for Seasonal Auto Regressive Integrated Moving Average and is an integrated model for non-stationary data. It has the parameters (p,d,q)(P,D,Q)S, with the "p" standing for the autoregressive model (Large "P" being the season AR) and the "q" for the moving average model("Q" is Seasonal MA) leaving the "d" and "D"(seasonal) standing for the differencing operator used. When working with any model the first part of the process is creating train and test data, for my timeseries I used the data up to the year 2010 as the train and the rest as test. The decision of the parameters started with differencing the train data and plotting the ACF and PACF. Since the only stationary process needed to be done was a seasonal differencing of ∇^{12} we can already assume that Seasonal differencing is 1 (D=1) and the other is 0 (d=0). From the PACF we can see a decay in seasonal lags meaning Seasonal MA (Q=1), the initial spikes being similar in both ACF and PACF, and the ACF having slight non-seasonal decay also leads us to believe that Non-seasonal ARMA can be p=2, p=3 or q=2. So the first 3 models I decided to produce was (2,0,0)(0,1,1)12 (Model1), (0,0,2)(0,1,1)12 (Model2) and (3,0,0)(0,1,1)12 (Model5). The other 2 models were calculated using auto.arima choosing the models with the lowest AIC. The models obtained were (0,0,10)(0,1,3)12 (Model3) and (0,0,3)(2,1,3)12 (Model4).

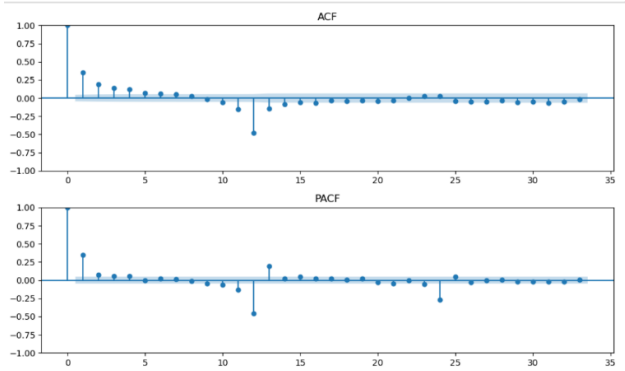


Figure 8. ACF and PACF of the differentiated Time Series

The Akaike Information Criterion (AIC) and the Bayesian information Criterion (BIC) are statistical measures that are used to compare the different models. They quantify the goodness of the fit and its complexity, with the lowest values generally being considered the best. When comparing all of the models AIC and BIC values, the model with the lowest AIC was Model 3 while the model with the lowest BIC was Model 5.

Table 2. AIC and BIC values of the 5 SARIMA models considered

Models	AIC	BIC
(2,0,0)(0,1,1)12	1340.70	1362.92
(0,0,2)(0,1,1)12	1394.39	1416.60
(0,0,10)(0,1,3)12	1311.80	1389.56
(0,0,3)(2,1,3)12	1371.72	1421.71
(3,0,0)(0,1,1)12	1327.59	1355.36

The forecasting was done using SARIMAX and the accuracy was calculated by doing a step-forward prediction. This prediction was then compared with the test by calculating the mean absolute error (MAE), the mean squared error (MSE), the root mean squared error (RMSE), the mean absolute percentage error (MAPE) and mean absolute scaled error (MASE) values manually. Looking at the table of errors the models got near identical results, this is also shown in the plots where all models have a good performance in comparison with the test data and are also very similar. In terms of errors the best performing model is model 2 containing the lowest errors in almost all metrics.

Table 3. Error values of the 5 SARIMA models considered

	MAPE	MASE	MAE	MSE	RMSE
(2,0,0)(0,1,1)12	4.45	0.17	0.33	0.14	0.38
(0,0,2)(0,1,1)12	4.18	0.15	0.30	0.12	0.35
(0,0,10)(0,1,3)12	4.68	0.18	0.36	0.17	0.41
(0,0,3)(2,1,3)12	4.35	0.16	0.31	0.13	0.36
(3,0,0)(0,1,1)12	4.54	0.18	0.34	0.15	0.39

After the analysis of both the errors and the AIC and BIC values, I considered model 5 the best performing model. This was decided due to all the results in both tables being very similar and the AIC and BIC values being considered more relevant. The diagnostics of model 5 shows no obvious patterns in residuals and has a KDE curve very similar to the normal distribution. The QQ plot shows that most of the points lie on the straight line even if the ends are slightly different and lastly 95% of correlations in the ACF of residuals are within the confidence band, showing us that it is a good model.

Model 5: (3,0,0) X (0,1,1)12

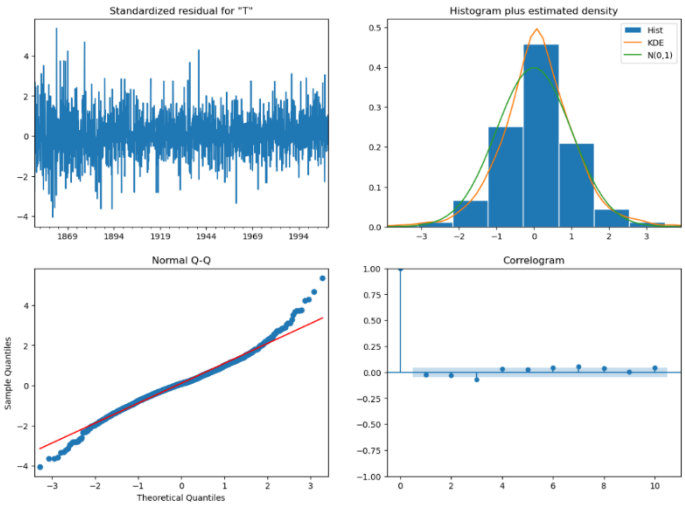


Figure 9. Diagnostics of SARIMA model (3,0,0)(0,1,1)12

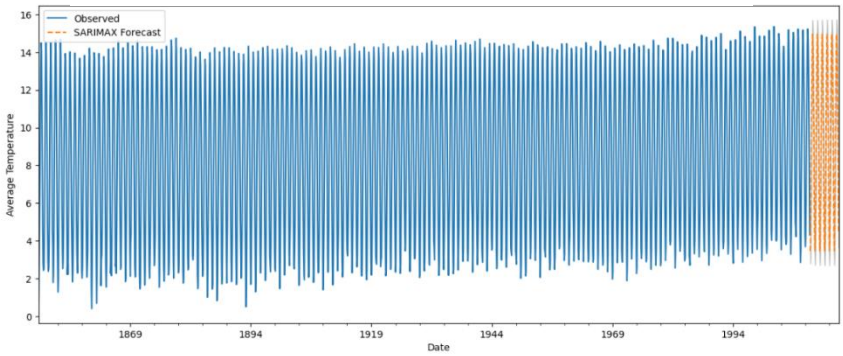


Figure 10. In point forecasting using SARIMA model (3,0,0)(0,1,1)12

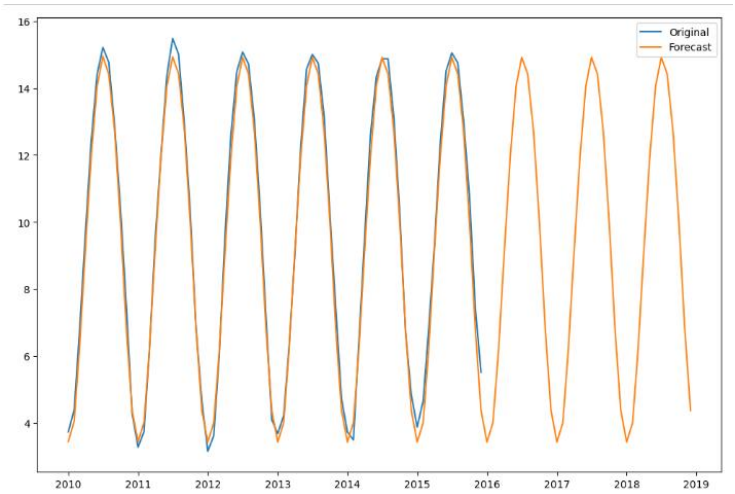


Figure 11. Out of point forecasting using SARIMA model (3,0,0)(0,1,1)12

Exponential Smoothing

Exponential smoothing is a forecasting method that uses weighted averages with the recent observations having higher associated weights. For time series that contain trend and seasonality the use of the Holt-Winter method is recommended. This method has two

variations: Additive and Multiplicative. The additive method being preferred when the seasonal variations are constant through the series and the multiplicative method being preferred when there is seasonality and trend. The plots for both methods show similar results, in terms of the errors the method with the lowest error metrics was the multiplicative method. This is not surprising since the decomposition of the time series showed an added trend to the strong seasonality, although the distinction of methods was almost inconsequential.

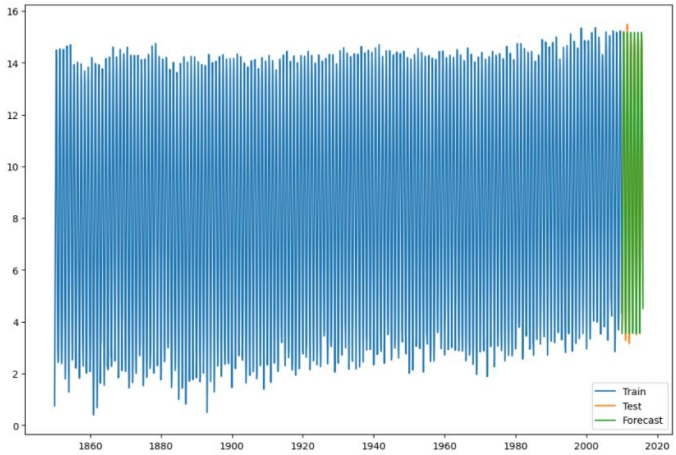


Figure 12. In point forecasting using Exponential Smoothing (Multiplicative)

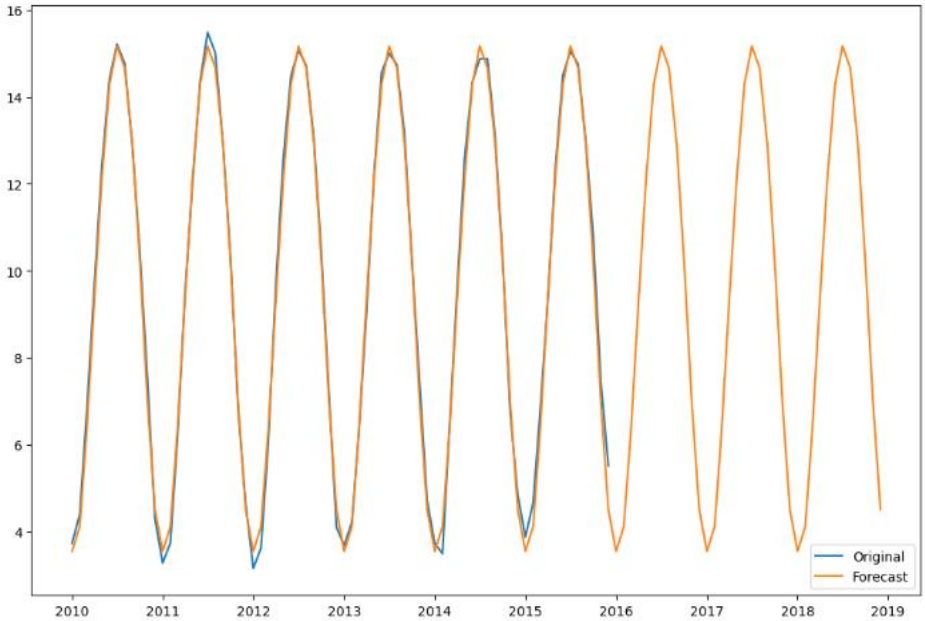


Figure 13. Out of point forecasting using Exponential Smoothing (Multiplicative)

Table 4. Error values of the Exponential Smoothing models

	MAPE	MASE	MAE	MSE	RMSE
Additive	3.82	0.14	0.26	0.11	0.31
Multiplicative	3.71	0.12	0.24	0.09	0.29

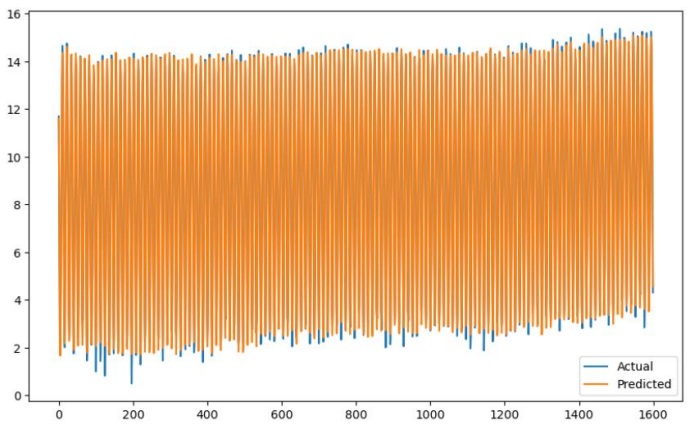
Cross-Validation

Cross-validation in time series is advantageous since it gives us a more robust model and reduces the chance of overfitting. CV in time series performs rolling-forecast origin in which you start with a subset of data for training and use the next few observations as the test set. After each test set you add the observations to the train set and repeat the process. The forecast accuracy is then calculated as the average of the errors. CV was used in all 5 SARIMA models and the exponential smoothing models, the mean squared error was then calculated to get the most accurate model. Looking at the table of errors, the model with lowest MSE was SARIMA model 3 ((0,0,10)(0,1,3)12), although all the CV instances had similar results with the difference being minimal (max difference of ± 0.05).

Table 5. Mean squared errors of the models used in Cross-validation

Model	MSE
(2,0,0)(0,1,1)12	0.100
(0,0,2)(0,1,1)12	0.104
(0,0,10)(0,1,3)12	0.097
(0,0,3)(2,1,3)12	0.102
(3,0,0)(0,1,1)12	0.099
Additive	0.100
Multiplicative	0.121

Figure 14. Forecasting using Cross-validation with SARIMA model (3,0,0)(0,1,1)12



Conclusion

In conclusion, the exploratory analysis of the time series showed strong yearly seasonality and a slight upward trend. The forecasting methods utilized showed similar results, which after evaluating each model the results showed high accuracy with low error values. In terms of the SARIMA models, the model (3,0,0)(0,1,1)12 was considered the best performing model since it had the lowest BIC value combined with being in the lower-end of the spectrum in terms of errors calculated. The exponential smoothing forecasting showed slightly smaller MSE values than the SARIMA models. Finally, a cross-validation of the time series was done to all forecasting

models in order to get a more robust and accurate model. This gave contradictory results since although the Cross-validation of the SARIMA models gave slightly lower errors, the CV in exponential smoothing had no effect or in the more specific case of multiplicative ES gave higher errors.

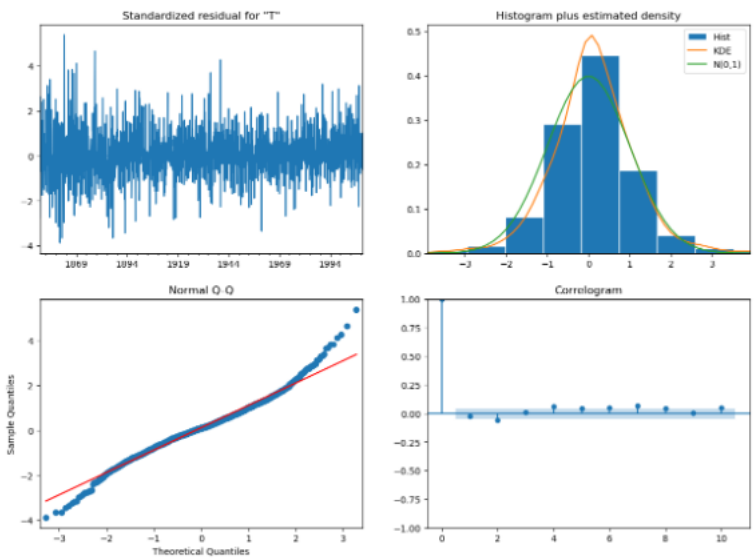
References

1. Berkeley Earth. Climate Change: Earth Surface Temperature Data. URL: https://www.kaggle. com/datasets/berkeleyearth/climate-change-earth-surface-temperaturedata/data.

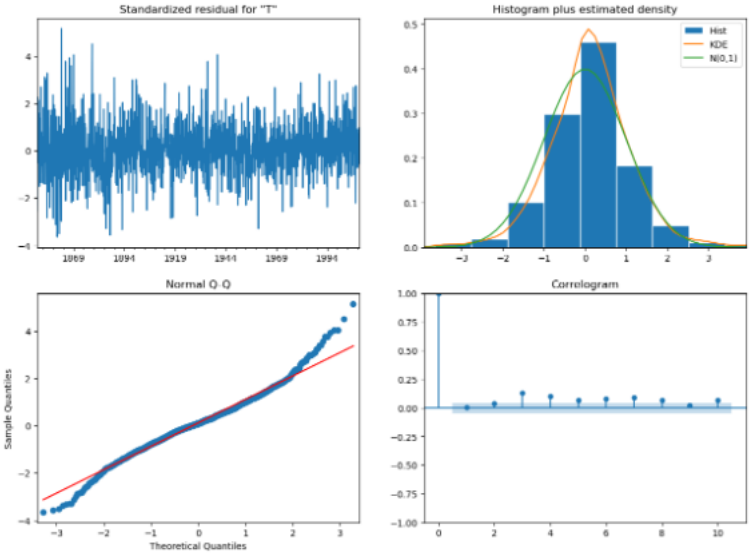
Attachments

SARIMA Models Diagnostics

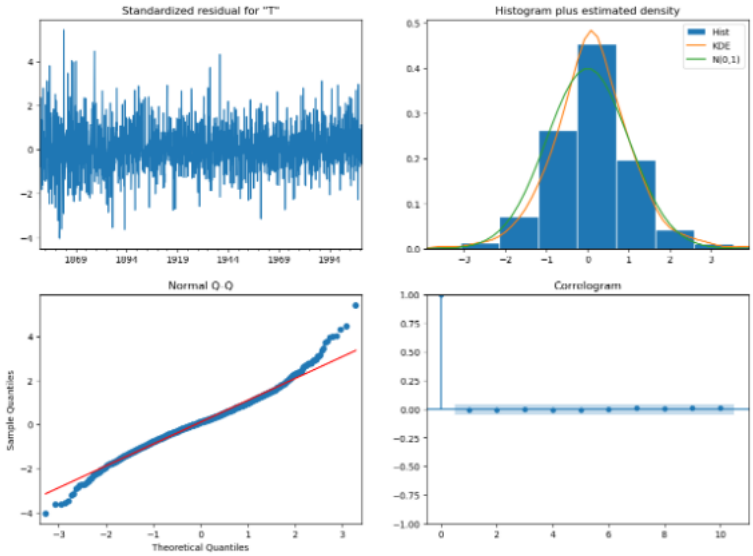
Model 1: (2,0,0) X (0,1,1)12

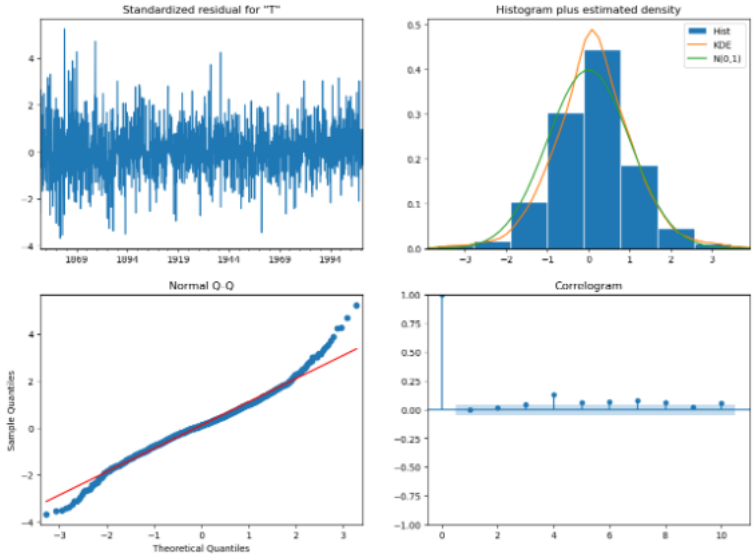


Model 2: (0,0,2) X (0,1,1)12



Model 3: (0,0,10) X (0,1,3)12



~

Model 4: (0,0,3) X (2,1,3)12